Tasbid Al Rahman

Dhaka, Bangladesh | +880 1780 824643 | tasbid-rahman.netlify.app | github.com/tasbidrahman10

RESEARCH INTERESTS

Security, robustness, and empirical evaluation of machine learning systems. My current research addresses intrusion detection in distributed training pipelines, alongside fine-tuning and evaluating a multimodal speech LLM. I am interested in extending this work toward the robustness, evaluation, and security of increasingly capable and agentic AI systems.

AI security · Robustness and anomaly detection · AI evaluations · Distributed and federated learning · Multimodal and agentic LLM systems

RESEARCH EXPERIENCE

Asynchronous Split Federated Learning for Intrusion Detection *In progress*

Undergraduate Researcher · Multi-university collaboration: North South University, BUET, KUET

- **Problem.** Intrusion detection across distributed client devices that cannot share raw data, where asynchronous participation makes the global model harder to keep both accurate and robust.
- **Method.** A privacy-preserving framework pairing split learning with asynchronous federated aggregation: clients train jointly while exchanging only intermediate activations with the server, so raw data stays on-device throughout training.
- **My contribution.** Direct the research and experimental workflow — dataset preparation, baseline models, and the evaluation plan spanning accuracy, communication cost, and robustness.
- **Status.** Manuscript complete; being prepared for submission to *Machine Learning* (Springer).

Python · PyTorch · Federated and split learning · scikit-learn

Empathetic Full-Duplex Speech LLM (EFSM) *In progress*

Deep Learning Research (CSE465) · North South University

- **Problem and method.** Make real-time spoken responses measurably more empathetic, using a fine-tuning setup built around Qwen2.5-Omni-7B with QLoRA at 4-bit precision.
- **Training design.** Freeze the Whisper encoder and the speech decoder and adapt only the language backbone, restricting adaptation to the component that generates response content and keeping the work feasible on limited hardware.
- **Evaluation design.** Specified a three-part protocol combining automatic and human measurement: EPITOME empathy scoring (0–6), word error rate on IEMOCAP and RAVDESS, and a human rating study across 20 audio samples.

Python · Qwen2.5-Omni-7B · QLoRA · Whisper · Gradio · EPITOME

MANUSCRIPTS

Asynchronous Split Federated Learning for Intrusion Detection — with collaborators at North South University, BUET, and KUET. Manuscript complete; submission to *Machine Learning* (Springer) in preparation — not yet submitted or published.

EDUCATION

North South University, Dhaka *2022 – present*

BSc in Computer Science and Engineering · Fourth year (final year)

Relevant coursework: Deep Learning (CSE465), Junior Capstone (CSE299), Software Engineering (CSE327).

SELECTED TECHNICAL PROJECTS

Smart Patient Follow-Up and Health Trend Analytics *Junior Capstone (CSE299)*

- Clinic follow-up system that schedules appointments and sends email and SMS reminders through n8n workflows, classifies patients as Critical, Moderate, or Normal from their vitals, and generates downloadable PDF health reports for providers.

Node.js · Express.js · MySQL · n8n · JWT

Zoco — Full-Stack E-Commerce Platform *Personal project*

- Storefront and admin dashboard covering size-variant inventory, order tracking, and sales analytics, with authentication, server-side rendering, and image hosting; deployed on Vercel and Supabase.

Next.js · TypeScript · PostgreSQL · Prisma · NextAuth.js · Tailwind CSS

TECHNICAL SKILLS

Machine learning: PyTorch, scikit-learn, federated and split learning, LLM fine-tuning (QLoRA, 4-bit quantization)

Languages and data: Python, TypeScript, JavaScript; NumPy, Pandas, Matplotlib

Systems and tooling: Next.js, Node.js, Express.js, PostgreSQL, MySQL, Prisma, Supabase, Git, n8n